

Production: Model Deployment

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$ echo "Data Sciences Institute"
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Introduction

Agenda

6.1 Model Deployment and Prediction Service

- ML Deployment Myths and Anti-Patterns
- Batch Prediction vs Online Prediction

6.2 Explainability Methods

- Partial Dependence Plots
- Permutation Importance
- SHAP Values

About

- These notes are based on Chapter 7 of *Designing Machine Learning Systems*, by Chip Huyen.

Our Reference Architecture

The Flock Reference Architecture

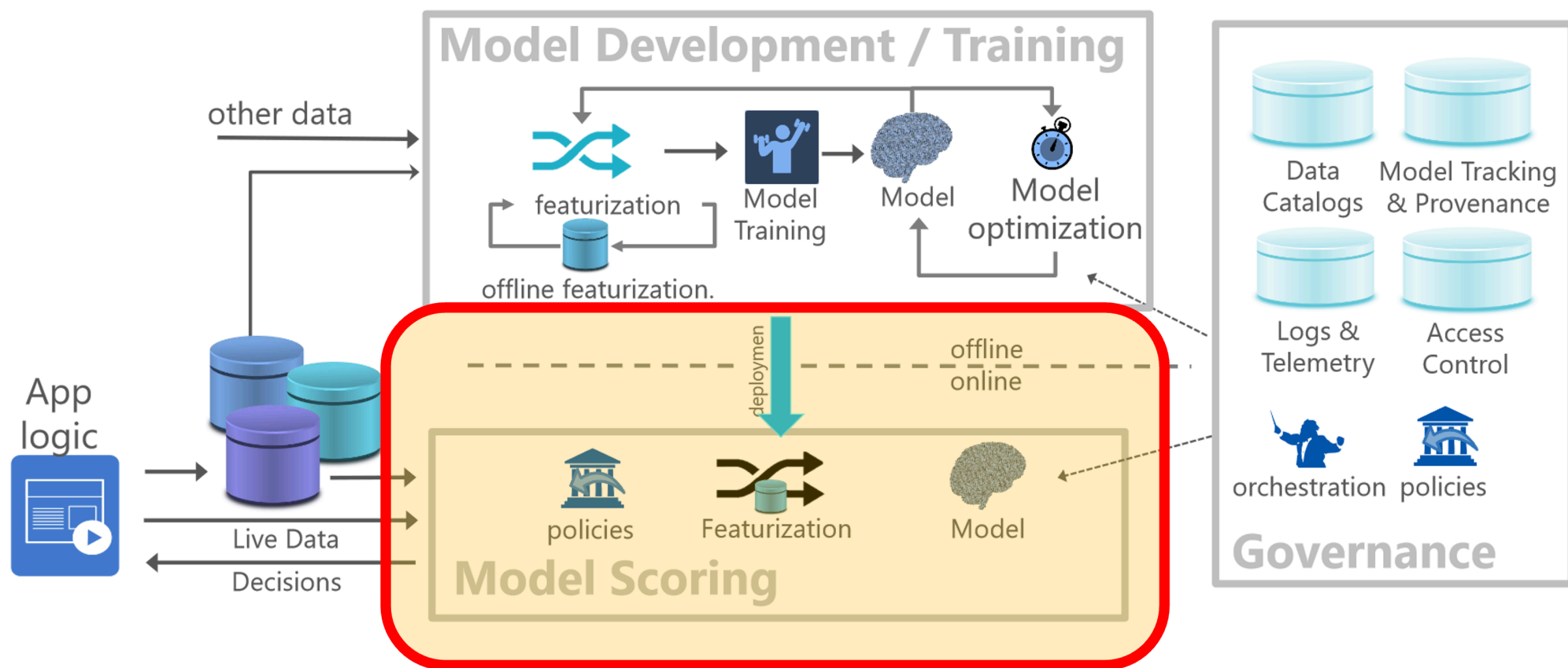


Figure 1: Flock reference architecture for a canonical data science lifecycle.

Deployment

Deployment

- Deploying a model means making it usable by allowing users to interact with it through an app or through a data product (BI visuals, reports, data views).
- Deployment is a transition of development to a production environment.
- There is a wide range of production environments, from BI to live applications serving millions of users.
- Engaging with users in formal or informal feedback conversations is helpful, although not always possible.

Deployment Myths and Anti-Patterns (1/3)

1. You only deploy one or two ML models at a time

- Infrastructure should support many models, not only a few.
- Many models can interact, and we also need a way of mapping these interactions.
- Ride-sharing app:
 - 10 models: ride demand, driver availability, estimated time of arrival, dynamic pricing, fraud, churn, etc.
 - 20 countries.

Deployment Myths and Anti-Patterns (2/3)

2. You won't need to update your models as much

- Model performance decays over time.
- Deployments should be easy:
 - The development environment should resemble the production environment as closely as possible.
 - Infrastructure should be easier to rebuild than to repair.
 - Small incremental and frequent changes.

Deployment Myths and Anti-Patterns (3/3)

3. If we don't do anything, model performance stays the same

- Software does not age like fine wine.
- Data distribution shifts: when the data distribution seen during training differs from the distribution encountered in production.

4. Most ML engineers don't need to worry about scale

- Scale means different things to different applications.
- Number of users, availability, speed, or volume of data.

Batch Prediction vs. Online Prediction

Online Prediction

- Predictions are generated and returned as soon as requests for these predictions arrive.
- Traditionally, requests are made to a prediction service via a RESTful API. When requests are made via HTTP, online prediction is known as *synchronous prediction*.
- Also known as on-demand prediction.

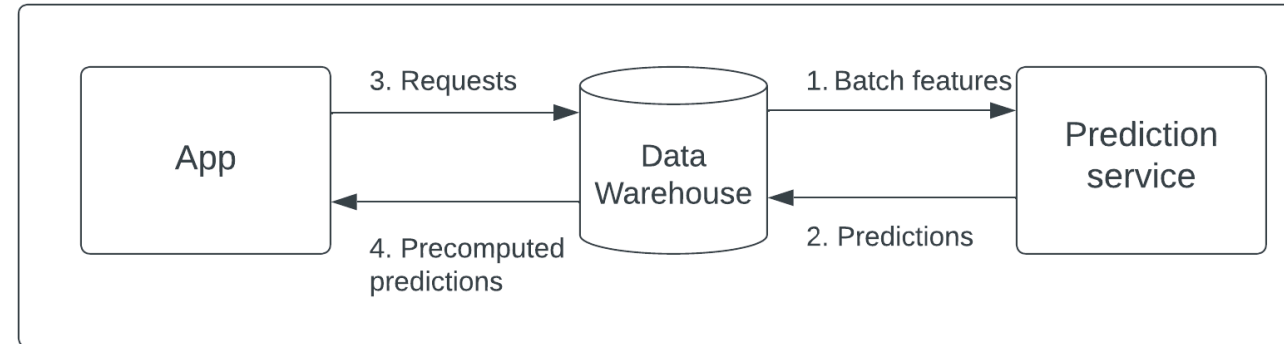
Batch Prediction

- Predictions are generated periodically or whenever triggered.
- Predictions are stored in SQL tables or in memory. They are later retrieved as needed.
- Batch prediction is also known as asynchronous prediction.

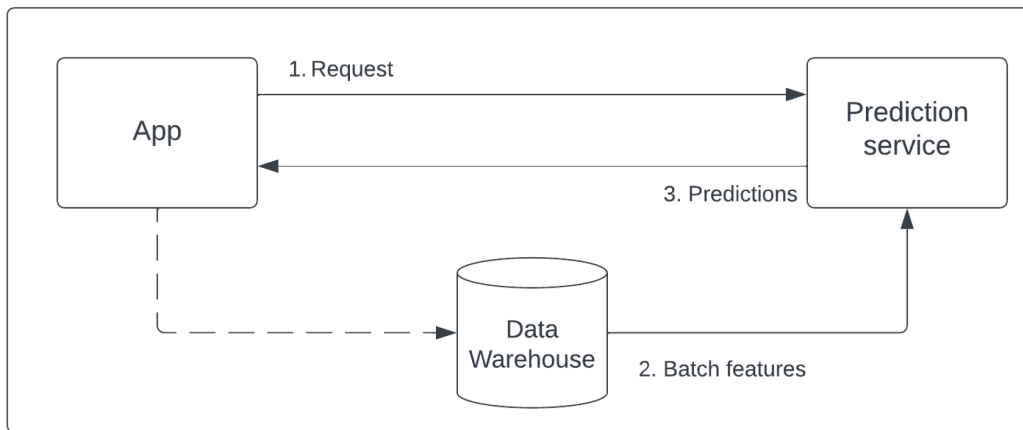
Model Prediction Service

Three types of prediction service:

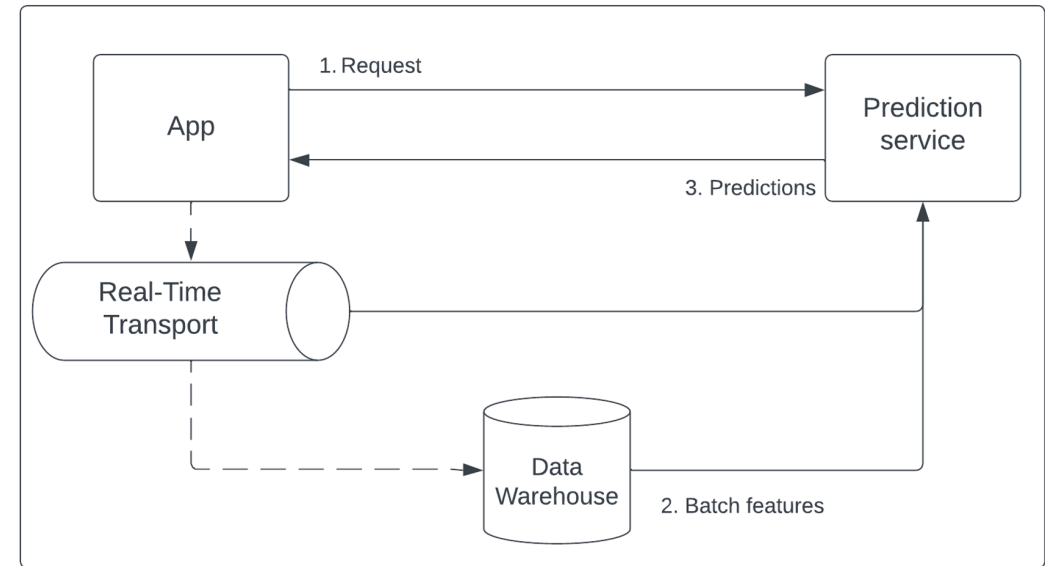
- Batch prediction: uses only batch features.
- Online prediction that uses only batch features (e.g., precomputed embeddings).
- Online streaming prediction: uses batch features and streaming features.



Model Prediction Service (cont.)



Online Prediction (based on Huyen 2021)



Streaming Prediction (based on Huyen 2021)

References

- Agrawal, A. et al. "Cloudy with high chance of DBMS: A 10-year prediction for Enterprise-Grade ML." arXiv preprint arXiv:1909.00084 (2019).
- Huyen, Chip. "Designing machine learning systems." O'Reilly Media, Inc. (2022).